Statement of Work

Optical Communications Telescope Dome

NASA Glenn Research Center

1.0 Introduction/Background:

The NASA Glenn Research Center (GRC) has a requirement for a telescope dome as part of its continuing development efforts in free-space optical and quantum communications systems in support of aeronatics and space operations missions. GRC is developing both transmit and receive optical systems and requires an enclosure to provide elemental protection from rain, snow, wind, etc., that will enable rooftop installation of a telescope.

2.0 Scope of Work:

The minimum technical requirements are:

- The dome must be a fixed, non-rotational, clamshell shutter design, providing a full 360° hemispherical view when fully opened.
- The opening and closing of the dome must be motorized, and open from the center with independent control of the shutters on each side. Fully indepedent actuation of each individual shutter is not required.
- A non-motorized manual capability to open and close the dome must be included.
- The dome must be wind and weather resilient when fully closed, providing protection from rain, snow, wind, sun, etc.
- The interior clear diameter must be ≥ 8.5 ft.
- The exterior diamter when fully opened must not exceed 12 ft.
- The interior height must be ≥ 8 ft, which may include base riser extensions.
- An access hatch in the base must be included.
- The dome must provide a feed-thru panel or other accomodation to allow for electrical, fiber-optic, ethernet, or other types of connections.
- Electrical must be 120 or 220 V single phase.

3.0 Delivery:

- The contractor shall deliver within 12 weeks after receipt of order to:

  NASA Glenn Research Center
  Bldg. 330
  21000 Brookpark Road
  Cleveland, OH 44135